

Cole A. Smith

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EDUCATION

University of Virginia – School of Engineering and Applied Science

Charlottesville, VA

Major: Mechanical Engineering – Rodman Scholar (*awarded to top 5% of engineering class*)

May 2026

GPA: 3.4

ENGINEERING EXPERIENCE

Solar Car Team at UVA

Charlottesville, VA

Brakes & Steering Lead; Fmr. Member, Brakes and Steering and Embedded Teams

August 2022 – June 2026

- Led full-cycle engineering, manufacturing, and team operations as Brakes and Steering lead; ran design reviews, onboarded new members, and executed root-cause analysis to completely resolve previous mechanical stability failures for competition
- Modeled dynamic steering geometry and brake line pressures; developed custom adjustable steering shaft, ratcheting handbrake, and steering arms; executed mid-race suspension retrofit and structural reinforcement of the steering shaft
- Directed a 5-week emergency sprint to manufacture the vehicle's brakes, steering, and wheel-end assemblies from scratch, preventing disqualification and delivering the car to competition
- Designed an embedded driver heads-up display using Raspberry Pi and STM32; integrated CAN bus and LoRa for wireless telemetry and developed a custom ReactJS dashboard

Polymer and Composites Research Group

Charlottesville, VA

Undergraduate Researcher

June 2025 – August 2025

- Fabricated flexible temperature sensors from a PEDOT:PSS/Carbon Nanotube (CNT) composite, investigating the impact of both CNT-to-polymer ratio and serpentine trace geometry on improving thermal sensitivity
- Developed a characterization testbed using a National Instruments (NI) DAQ to analyze the stability of the sensor's resistance under hydrodynamic conditions; presented findings at a research symposium

Commonwealth Center for Manufacturing (CCAM)

Disputanta, VA

Manufacturing Processes Intern

May 2024 – August 2024

- Conceptualized and led an intern capstone project for a UR5 collaborative robot, overseeing timeline management and final demonstration to senior staff to enable automatic reorientation for heavy, cumbersome manufacturing assembly processes
- Engineered a custom pneumatic gripper and its full actuation system from concept to completion, handling CAD design, fabrication (Wire EDM), and integration with UR5 controls to enable vision and gesture-driven automatic part handling
- Developed and benchmarked two competing personnel tracking systems; one vision-based using ArUco tags and a second using custom ultra-wideband radio modules, achieving 10cm precision at 5% the cost of commercial options

Lightstorm Research

Charlottesville, VA

Summer Intern

June 2022 – August 2022

- Designed and programmed a complete user interface for RP2040-based custom sensing units in MicroPython, integrating sensors, LCD display, user controls, memory, and wireless telemetry; developed functional breadboarded prototypes
- Assembled, reflow soldered, microscopically inspected, and flashed 50+ PCBs to validate sensor modules
- Implemented register-level sensorless stepper motor homing using current feedback to achieve closed-loop positioning for precision movement
- Diagnosed and rebuilt a custom circuit after thermal failure, disassembling, labeling, and replacing key components (Intel NUC boards) to restore reliable system operation

SKILLS AND CERTIFICATIONS

Design and Analysis: SolidWorks CAD (CSWA) and FEA, Siemens NX, Fusion 360, MATLAB, VBA

Fabrication: CNC Machining (CAM), Lathe, Mill, Waterjet, 3D Printing (FDM), PCB Assembly & Rework (SMT, Reflow, Hot Air)

Electronics and Software: PCB Design (KiCAD), STM32, ESP32, C++, Python (NumPy, Pandas, OpenCV), Java

Leadership and Activities: Eagle Scout, President of Phi Sigma Kappa, Jazz and Classical Flautist, Soloist for UVA Jazz Ensemble